

Spirometer Project Progress Report

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Project focus: Gamified incentive spirometer for pediatric respiratory therapy.
Main platform: Raspberry Pi 4, 3.5-inch LCD touch screen, flow sensor, Python

Summary:

- Built the software side end to end, from sensor readout to finished gameplay.
- Implemented sensor driver, filtering, smoothing, validation screen, and calibration.
- Used the calibration formula $\text{Flow (L/min)} = \max(0, 0.503 \times \text{Hz} - 29.70)$.
- Added therapy session logic, scoring, data logging, settings, and menu-after-battery behavior.
- Pivoted from a Flappy Bird style game to a car-based game for steadier exhalation therapy.

For the full detailed report, contact Abdulbosir Akhmadjonov.